

Global Logistics Business Intelligence & Analytics

End-to-End Logistics BI Project | Python • PostgreSQL • SQL • Tableau

1. Project Overview

This project is an end-to-end Business Intelligence solution for analyzing global logistics operations. It transforms shipment, customer, product, route, carrier, driver, vehicle, warehouse, and weather data into actionable insights through a centralized data warehouse and interactive Tableau dashboards.

2. Business Objectives

- Monitor shipment and delivery performance.
- Analyze operational delays and transit efficiency.
- Evaluate logistics costs and financial performance.
- Understand customer revenue and segmentation.
- Analyze driver and vehicle performance.
- Measure the impact of weather and road conditions on shipments.
- Identify high-volume and high-delay routes.
- Support data-driven operational decision-making.

3. Data Architecture

The warehouse follows an Enterprise Star Schema approach. Dimension tables provide descriptive business context while fact tables store measurable logistics events and performance metrics. The model was designed to support scalable SQL analysis and Tableau reporting.

Table	Type	Purpose
fact_shipments	Fact	Central shipment-level operational measurements.
dim_date	Dimension	Calendar and time attributes.
dim_customers	Dimension	Customer attributes and segmentation.
dim_products	Dimension	Product attributes.
dim_routes	Dimension	Origin, destination, route, and transport attributes.
dim_carriers	Dimension	Carrier attributes.
dim_drivers	Dimension	Driver attributes and performance measures.
dim_vehicles	Dimension	Vehicle attributes and capacity/efficiency measures.
dim_warehouses	Dimension	Warehouse attributes.
dim_weather	Dimension	Weather severity, condition, and road attributes.

4. Data Pipeline

Source data is prepared and transformed using Python and Pandas, loaded into PostgreSQL, organized into a dimensional warehouse model, validated with SQL, and consumed by Tableau for interactive reporting.

Source Data → Python/Pandas → Cleaning & Transformation → PostgreSQL → Star Schema → SQL Validation → Tableau

5. Technology Stack

Technology	Role
Python / Pandas	Data preparation, generation, cleaning, and transformation
PostgreSQL	Central analytical data warehouse
SQL	Analysis, validation, and business logic
Tableau	Interactive BI dashboards and visualization
Git / GitHub	Version control and portfolio documentation

6. Tableau Dashboard Suite

- 1. Executive Overview** — High-level logistics KPIs and overall business performance.
- 2. Operations Performance** — Shipment status, delivery performance, delays, carriers, and operational trends.
- 3. Financial & Cost Analysis** — Revenue, logistics costs, and financial performance.
- 4. Customer Revenue & Segmentation** — Customer contribution, revenue distribution, and segmentation.
- 5. Driver & Vehicle Performance** — Driver score, experience, vehicle efficiency, utilization, and fleet metrics.
- 6. Weather & Shipment Impact** — Weather severity, road conditions, and their impact on shipment delays.
- 7. Route Performance & Network** — Route volume, average delays, geographic distribution, and shipment-volume relationships.

7. Key Business Questions

- Which routes handle the highest shipment volumes?
- Which routes experience the highest average delays?
- Which carriers and transport modes perform most effectively?
- How does weather severity affect shipment delays?
- Which drivers and vehicles perform most efficiently?
- Which customer segments contribute the most revenue?
- Where are operational bottlenecks concentrated?
- Is shipment volume associated with average delay?

8. BI & Analytics Capabilities Demonstrated

- Enterprise Star Schema and dimensional modeling
- Data transformation and ETL concepts
- SQL-based analysis and validation
- KPI design and business metric definition

- Interactive Tableau dashboard development
- Geographic and route network analysis
- Customer segmentation and revenue analysis
- Fleet, driver, and vehicle analytics
- Weather and operational risk analysis
- Data quality and final dashboard QA

9. Data Quality & Validation

The project included schema validation, relationship checks, calculated-field testing, dashboard-level QA, visual consistency checks, and review of KPI definitions before finalizing the Tableau dashboard suite.

10. Future Improvements

- Real-time shipment tracking and operational monitoring.
- Predictive delay and ETA modeling.
- Machine-learning-based route and delivery optimization.
- Automated data pipelines and scheduled warehouse refreshes.
- Cloud deployment and scalable data processing.
- Automated Tableau refresh and alerting workflows.

11. Project Status

Completed: data warehouse architecture, dimension and fact modeling, data transformation workflow, SQL analysis and validation, seven Tableau dashboards, and final dashboard QA.

12. Author

Mohamed Alneel

Business Intelligence & Data Analytics

Python • SQL • PostgreSQL • Tableau • Excel